

De Montfort University

Beng (Hons) Mechanical Engineering – Year 3

Objective: Apply a Total Design Process from User Need to Concept Selection

Coursework: Product Design & Materials Selection

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Abstract

Kitchen nutcrackers can be designed for display or efficient use in a kitchen. This report demonstrates the process of finding the material and general design of a nutcracker for kitchen use. An effective nutcracker for the average household must be reasonably priced, last as long as other kitchenware and be efficient to use. These criteria require step-by-step screening to remove materials and designs that are not fit for this purpose. A nutcracker should be ergonomic, quick and light as an effective consumer product. The best design that fits these criteria is a hinge nutcracker which utilises leverage to crack hard nuts; this is easy and intuitive to use. The material which is best suited for this product scope is annealed stainless steel 304. It is £3/kg which is affordable, can be manufactured using investment casting, has a young's modulus of above 10 GPa and a yield strength of over 100 MPa making it fit to handle hard nuts like macadamia nuts.

What Product:

Nutcracker

Market/User need & PDS:

Need:

People that enjoy or need to crack their nuts, desire an easy, quick, and fatigue-free to use tool to break open their nuts.

Key stakeholders:

Chefs, nut-eaters.

Function

Multiply user force input, to crack nut.

Performance

Safety factor ≥ 2 .

User/Market

Ergonomic handles. Grippy handle. Reasonable size.

Cost/Manufacturing

Cost of target units. Ability for round shapes.

Standards/Safety

Fit in average kitchen drawer. Hangable kitchen utensil. Safe edges. Food-safe, bacteria.

Environment/Lifecycle

Recyclability or low embodied energy preferred. Corrosion resistance. Dishwasher safe.

- Define user or market need
- Must be measurable such as weight
- Requirements should not be solution bias: avoid pre-committing to forms/materials

Concept Generation

Function structure:

User force → force transmits to nut → nut cracks.

adjustable cracker size.

1. Cracker type: screw nutcracker, hinge nutcracker
2. Main material: stainless steel, aluminium, wood
3. Grip strategy: dishwasher-safe rubber, textured grooves

Idea	Cracker type	Main material	Grip strategy	Review
C1	Screw nutcracker	wood	Textured grooves	Not fast and easy, must screw down to crack and unscrew after each nut does not meet PDS.
C2	Hinge	wood	rubber	Aesthetic, light, strong. Fast and easy at cracking due to hinge. Rubber may not meet aesthetics of wood. Raw wood attracts and traps bacteria
C3	Hinge	Wood	Textured	Same as 2 nd with removal of odd aesthetic of rubber and wood.
C4	Hinge	Anodized aluminium	rubber	Lightweight. Not as corrosion resistant as steel, may fail to fatigue. Rubber is ergonomic
C5	Hinge	Anodized aluminium	texture	Same as other aluminium. Texture is slightly less grippy
C6	Screw	Steel	rubber	Strong. Very corrosion resistant. Not ergonomic. Slow to use. Noisy.
C7	Hinge	steel	rubber	Moderate weight. Strong enough for all nuts. Does not trap bacteria. Dish washer safe. Rubber grip is very grippy and aesthetically pleasing with chrome finish of stainless steel.
C8	Hinge	steel	textured	Same as 5 th but slightly worse grip and aesthetics.

Pugh Matrix:

Criteria:

1. Fatigue Resistance
2. Mass

3. Cost
4. Strength
5. Manufacturability
6. Speed of use
7. Aesthetics
8. Ergonomics
9. Dishwasher safe
10. Bacteria
11. Corrosion resistance



Click to see full view






KitchenCraft Nutcracker, Metal, 16 cm

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3.8 ★★★★★ (578)

£5⁹⁹ (€0.50 / count)

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Colour: **Silver**

Size: **1.5 x 7 x 21.2 cm**

- This cracker makes it easy to enjoy your favourite nuts! It's a great-value tool that's designed to last
- It has 2 jaws to crack nuts of all types and sizes. You can even use them to crack open crab and lobster claws
- With its serrated jaw you don't have to worry about slips. This nut cracker securely grips shells as you squeeze
- Made of metal with a shiny chrome effect, so it can handle knocks and bumps and looks great at the Christmas dinner table!
- The nutcracker is dishwasher safe and comes with a 12 month guarantee

Additional details

Made in Italy

 This product was wholly produced or has undergone its last substantial transformation in Italy. [Discover Made in Italy](#). [Learn more](#)

Datum

Concept	stre.	Mass	Cost	Manu.	Spe.	Aest.	Ergo	Dish.	Bact.	Corr.	T
C1 (screw/wood)	S	+	-	-	-	+	-	S	-	S	-3
C2 (wood/rubber)	S	+	-	-	S	S	S	-	-	S	-3
C3 (wood/texture)	S	+	-	-	S	S	S	S	-	S	-2
C4 (alu/rubber)	--	+	+	S	S	S	+	-	S	-	-1
C5 (alu/texture)	--	+	+	+	S	S	S	-	S	-	0
C6 (steel/screw)	S	S	S	S	-	S	-	S	S	S	-2
C7 (steel/rubber)	S	S	S	-	S	+	+	-	S	S	0
C6 (steel/texture)	S	S	S	-	S	S	+	S	S	S	0

Selected Concept

Stainless steel, rubber grip pivoting nutcracker.

Reasoning:

- Excellent stiffness and fatigue resistance
- Aesthetically pleasing
- Better ergonomics and faster than screw nutcracker
- Extremely corrosion resistant.

Translation Step

Function:

Lever arm.

Constraints:

Length (L) fixed by ergonomics. Must not yield or fracture under the maximum human grip force or the maximum nut strength. Must be food safe and corrosion resistant.

Objective:

Minimise mass for ease of use and minimise cost for mass production.

Free variables:

Material choice, cross sectional thickness & shape.

Materials Index Derivation

To minimise mass at correct stiffness:

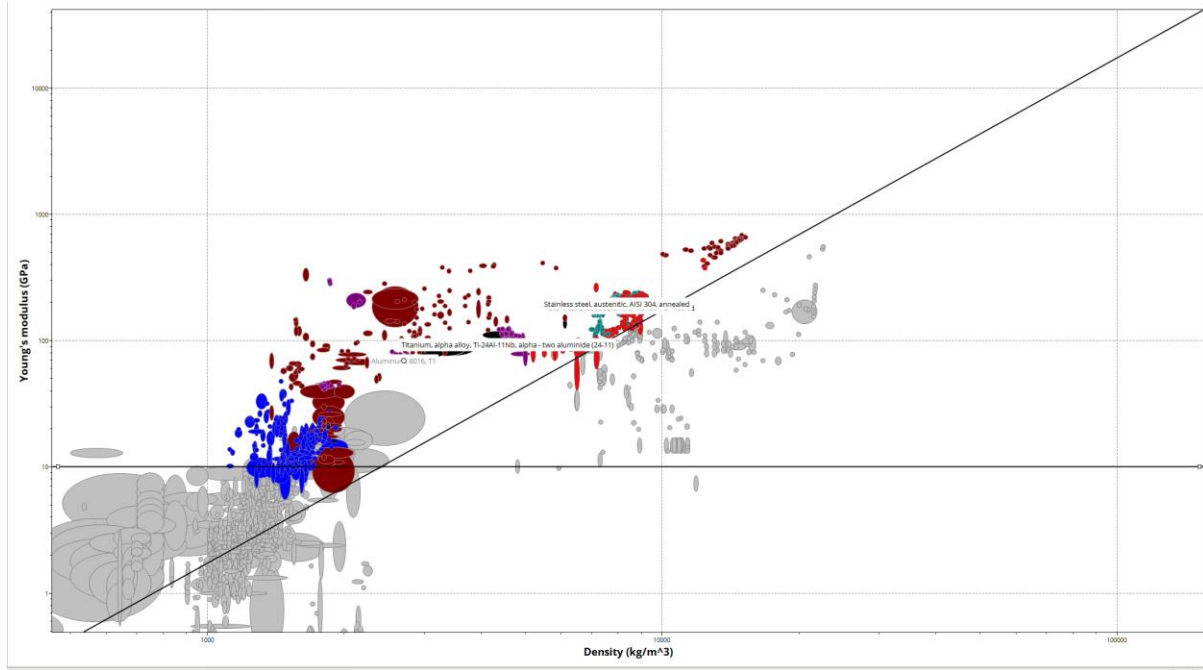
$$M_b = \frac{E^{\frac{1}{2}}}{\rho}$$

Higher M_b the better for stiffer, lightweight lever/beam.

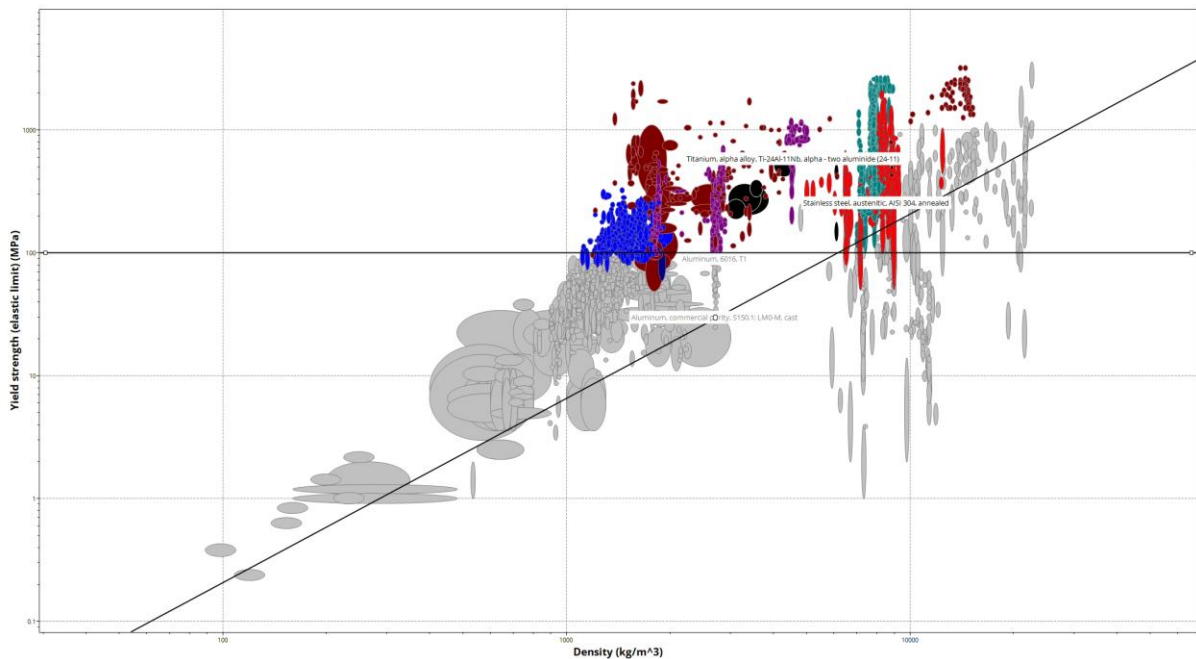
Screening

$\sigma_y > 100 \text{ MPa}$ the handles must not permanently bend or snap when cracking a hard nut (macadamia for example).

$E > 10 \text{ GPa}$ to ensure the force transfer from the user's hand goes to the nut and doesn't excessively transfer to bending the handles instead.



$M_b = \frac{1}{\rho} \frac{E^{\frac{1}{2}}}{\rho} = 2$. Bending is very sensitive to the thickness of the handle; to double the stiffness, only a small increase in thickness is needed. Therefore, a gradient of 2 used at approximately 1100 kg/m³ density against 10 GPa. 2 is high stiffness so that nutcracker works efficiently and doesn't bend excessively when used. All materials under 10 GPa young's modulus are not included.



The performance index $M = \frac{\sigma_y^{\frac{2}{3}}}{\rho}$ was used. Strength is the resistance to snapping and is less sensitive to a material's thickness to weight ratio than stiffness, so a 1.5 gradient is used. 1.5 line based on aluminium commercial purity for reasonable baseline in yield strength against density. Also, all materials below 100 MPa are not included.

Must be able to survive 5+ years against dishwashing and general washing via corrosion resistance.

Should be non-toxic (food grade), material must not leach chemicals.

Manufacturing should be compatible for mass production.

Ranking (qualitative)

1. Stainless steel (304/316): durable and food safe. High density, 100% dishwasher safe. Ashby chart showed it had yield strength > 100 MPa and young's modulus > 10 GPa. Price ≈ £3/kg, cheap price can fit into sale price of < £10. Manufactured using investment casting where molten steel is poured into ceramic moulds. This can provide complex geometries not normally possible with other methods such as drop forging. The mould is then trimmed of excess metal, drilled for the hinge holes and milled to sharpen any teeth. Until this point, the stainless steel has not yet been annealed, this is for easier manufacturing but, leaves the stainless-steel brittle, which is a problem for a tool such as a nutcracker where strong nuts can cause the tool to shatter. So, it is then annealed (heated to 1010°C -1120°C until crystalline structure relaxes, rapid quenching), which provides maximum corrosion resistance and its ability to bend under pressure slightly instead of snapping. Surface is then polished and product assembled. Overall best choice, fits market needs and is relatively cheap, can fit product specifications and price.
2. Cast aluminium alloys: high $\frac{E^{1/2}}{\rho}$ and low cost. Lightweight and stiff but can corrode in harsh dishwasher detergents over time. Yield strength < 100 MPa. Price ≈ £0.70/kg. extremely cheap. Lower melting point than stainless steel (about 660°C), so cheaper to melt. High-pressure die casting which is quicker since aluminium can cool faster in this process. Trimming of excess scrap and tumbling (parts put into vibrating container with stones to smooth out rough edges). Anodization and assembly. Overall cheapest manufacturing and material but does not fit user need.
3. Titanium alloys: exceptional stiffness to mass, food safe. Excessively high cost; only for luxury products, price ≈ £10/kg. CNC machining, assembled. Titanium has a special problem called galling where if two titanium surfaces rub together, they can cold-weld together, causing the hinge on the nutcracker to seize up. This is prevented by using a tungsten carbide coating or using a completely different material for the hinge pin (like stainless steel or bronze). Overall, most expensive and more fit for a high-grade nutcracker which is not the objective.

Stiffness check

Force ≈ 500 N

Length ≈ 0.15m

Target deflection (δ): <2 mm

Material (E): stainless steel = 200 GPa. Aluminium = 70 GPa.

$$\delta = \frac{FL^3}{3EI}$$

Rearrange to find:

$$I = \frac{FL^3}{3E\delta}$$

I is second moment of area. F is force applied; L is length of the lever. E is young's modulus of the material and δ is the deflection under load.

$$I = \frac{400 \times 0.15^3}{3 \times (200 \times 10^9) \times 0.002} \approx 1.13 \times 10^{-9} m^4$$

$$I = \frac{\pi D^4}{64}$$

Where D is the diameter.

$$D = \sqrt[4]{\frac{I \times 64}{\pi}} = 0.0123 \text{ m}$$

Risk & Sustainability

Risks:

Corrosion from dishwasher detergents (highly alkaline).

Fatigue cracking at pivot point, high stress at this point in a small area. Hundreds to thousands of repeated forces can cause failure from fatigue.

Stress concentrations can cause failure if geometry is not accounted for.

Sustainability:

Stainless steel is 100% recyclable at the end of its life, unlike many plastics which end up in landfills.

Aluminium is easier to recycle; stainless steel generally has a smaller carbon footprint during initial production especially when made from recycled scrap.

A nutcracker is generally a "built for life" item, this means the sustainability impact is not as bad as other products especially when they are built of plastics.

Conclusion

A hinge annealed stainless steel 304 using investment casting for manufacturing is the best choice to fit an average nutcracker that is efficient, lightweight and cheap. Aluminium and titanium are great materials and provide cheapness and detail respectively but do not fit the product criteria.